



Modular Athletic Exoskeleton Project
Power Architecture Testing, Sensor Validation, and PCB Progress Update

1) Power Architecture Testing (Single Supply Distribution vs. Per-Node Battery)

Objective

We performed extensive power testing to evaluate the feasibility of using one central battery/power supply and distributing power to all sensor nodes, versus using individual batteries per node. The goal was to understand real-world power loss, voltage drop behavior, and system reliability impacts in a wearable setup.

Test Setup

To simulate the expected deployment conditions, we built a test setup representing the complete system power path:

1. Main power source (single supply/battery equivalent) connected to the main controller board
2. Power distributed from the main board to multiple sensor nodes through wiring harness / interconnect cables
3. Each sensor node load included:
 1. IMU sensor load (active sampling)
 2. Microcontroller load
 3. Haptic motor load (burst current / transient events)
4. Power measurements taken at:
 1. Source output
 2. Main board input
 3. Node input points
 4. During both steady-state operation and dynamic load events (sensor sampling + haptic bursts)

This allowed us to observe worst-case behavior including transient current spikes and real cable losses.

Observations (Single Supply Distributed Power)

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Using a single power source and distributing power outward introduces measurable loss and reliability concerns, especially as wiring length and dynamic loads increase.

Key Cons / Limitations Identified

1. Voltage drop across wiring and connectors

- Distributed power introduces IR drop through cables and connector contact resistance.
- Nodes farthest from the source see the largest drop.

2. Power loss / wasted energy

- Cable resistance converts power into heat.
- Loss increases with current draw, especially during haptic events.

3. Haptic burst current impacts other nodes

- When one node triggers haptics, the sudden current spike can cause a brief voltage dip.
- This can affect neighboring nodes and cause unstable behavior in sensing/communication.

4. Risk of brownout / resets

- Under worst-case current draw, some nodes may drop below minimum operating voltage.
- This increases the risk of resets, sensor glitches, or data corruption.

5. Noise coupling through shared supply

- A shared rail allows noise from one node (motor switching, digital activity) to propagate across the network.
- This can degrade sensor performance or communication stability.

6. Scaling becomes difficult

- Adding more nodes increases total current and increases voltage drop and instability.
- This makes the system less scalable without heavier wiring or higher voltage distribution + regulation.

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7. Mechanical complexity

- Centralized power requires thicker power distribution wiring.
- Adds complexity for routing in wearable form factor and increases failure points.

Final Decision

Based on the results, we finalized that the system will be more efficient and reliable if each node has its own local battery.

Why per-node battery is better

- Eliminates long-distance voltage drop issues
- Removes shared-rail interference between nodes
- Improves stability during haptic bursts
- Better scalability (adding nodes does not overload a central rail)
- Simplifies wiring harness requirements (data lines only, or reduced power dependency)

This architecture is more aligned with a wearable system where consistent performance and robustness are required during real user motion and dynamic loads.

2) Sensor Validation Testing (Sports Expansion + IMU Saturation Check)

Objective: One key concern raised was whether the selected sensors (specifically the IMU) could saturate during real-world high-acceleration motion. To address this, we performed additional validation testing using the **previous sensor setup**, expanding beyond baseball to evaluate performance across **multiple sports and field activities**.

The goal was to confirm that the selected sensor stack can reliably capture motion without clipping, saturation, or unstable behavior during fast, high-dynamic movements.

Test Setup / Method

1. Used the previously validated sensor hardware setup (IMU + controller logging)
2. Tested motion profiles that represent different sports movements (higher dynamics, sharper direction changes, faster swings/impacts)

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3. Focus areas:

1. Confirm IMU measurement range is sufficient (no clipping/saturation)
2. Confirm stable sampling and consistent output under fast motion
3. Confirm sensor behavior remains reliable during repeated motion cycles

Results / Findings

- The expanded testing showed that the sensor selection remains suitable for dynamic motion use cases.
- The IMU behavior was monitored to ensure:
 - No obvious saturation artifacts during high-speed motion
 - Consistent data capture across repeated trials
 - This additional testing increases confidence that the sensor selection will support broader sports use cases and will not be limited only to one activity type.

3) Schematics and PCB Layout Area Summary

Status: Schematics work is completed. Below is the PCB area breakdown and recommended PCB size options for both the Main PCB and Sensor PCB based on component placement requirements.

Along with the area sizing, we also evaluated the tradeoffs between **single-sided assembly** and **double-sided assembly**, since this impacts **cost, mechanical stack-up, thickness, and manufacturability**.

Main PCB: Total Component Area: 1597.2 sq. mm

Single-sided Assembly

- Total PCB Area: 4563 sq. mm
- Possible PCB Dimensions:
 - 70mm × 70mm
 - 90mm × 55mm

Notes: This option is cost-effective for assembly and simpler mechanically, but requires a larger footprint.

Double-sided Assembly

- Total PCB Area: 2457 sq. mm
- Possible PCB Dimensions:

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- 55mm × 55mm
- 70mm × 40mm

Notes: This option reduces overall board area significantly, but increases assembly cost and enclosure stack-up constraints.

Sensor PCB: Total Component Area: 844.493 sq. mm

Single-sided Assembly

- Total PCB Area: 2413 sq. mm
- Possible PCB Dimensions:
 - 40mm × 40mm
 - 50mm × 30mm

Notes: Lower manufacturing complexity and easier rework, but footprint is larger.

Double-sided Assembly

- Total PCB Area: 1299 sq. mm
- Possible PCB Dimensions:
 - 50mm × 50mm
 - 60mm × 40mm

Notes: Smaller footprint and better for compact wearable placement, but increases assembly complexity and mechanical constraints.

Assembly Tradeoff Summary

Option A - Single-Sided Assembly (Components on one side)

Pros

- Lower assembly cost (single-side pick & place)
- Easier manufacturing and inspection
- Easier rework/debug during prototypes
- Generally reduces risk in early builds
- Can reduce overall PCB “height” on one side (cleaner mechanical stacking)

Cons

- Requires larger PCB area to fit all components
- Larger board may increase enclosure size or restrict wearable placement

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- Routing density may become a constraint as features expand

Option B - Double-Sided Assembly (Components on both sides)

Pros

- Smaller PCB footprint (more compact)
- Better for tight enclosure constraints / wearable form factor
- Allows denser placement and future expandability without growing board size

Cons

- Higher assembly cost (two-side pick & place + additional process steps)
- Increases mechanical stack-up complexity
(components exist on both sides → tighter height/clearance constraints)
- More complex to debug/rework compared to single-sided boards
- Higher risk of mechanical interference depending on enclosure design

Summary

- Power testing confirms that single-source distributed power introduces voltage drop, wasted power, and stability risks under dynamic loads.
- We finalized that per-node batteries provide a more stable and scalable solution for the wearable system.
- Additional sensor testing across multiple sports-like movements confirms the sensor selection remains suitable and reduces risk of IMU saturation concerns.
- Schematics are complete and PCB sizing options are finalized for both main and sensor boards.